No. H9D3061

TO:

# SPECIFICATION

DESCRIPTION : AXIAL LEAD CERAMIC CAPACITOR
PARTS NAME : 050 type
Rated Voltage : 16,25,50 V D.C

TAIYO YUDEN CO., LTD.
QUALITY ASSURANCE DEPT.

DATE : 29. AUG. 2003

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# 1. General

## 1.1 Scope

This specification applies to Axial Lead Ceramic Capacitors (called Capacitors hereafter) for general use in electronic equipment.

## 1.2 Quality

Capacitors are manufactured in a highly quality—controlled processes to ensure the reliability of the products.

## 1.3 Patents

Any problems related to the patents on the Capacitors shall be the responsibility of supplier.

# 2. Ordering Code

< Example >

| UP050 | SL | 150 | J | — | A | — | B |
|---|---|---|---|---|---|---|---|
| ① | ② | ③ | ④ | | ⑤ | | ⑥ |

① Type Designation
② Temperature Characteristics
③ Nominal Capacitance
④ Capacitance Tolerance Symbol
⑤ Classification Symbol
⑥ Packaging Symbol

## 2.1 Type Designation

Type Designation shall consist of five numerals.

| Code | Rated Voltage |
|---|---|
| UP050 | 50 V D.C |
| TP050 | 25 V D.C |
| EP050 | 16 V D.C |

## 2.2 Temperature Characteristics

Temperature Characteristics shall be expressed by English capital CH, RH, UJ, SL, B, X, Y, or F.

| Characteristics | Capacitance Temperature coefficient | Characteristics | Capacitance Temperature coefficient |
|---|---|---|---|
| CH | 0 ppm/℃±60 ppm/℃ | B | ±10 % |
| RH | −220 ppm/℃±60 ppm/℃ | X | ±15 % |
| UJ | −750 ppm/℃±120 ppm/℃ | Y | ±22 % |
| SL | +350 ppm/℃～−1000 ppm/℃ | F | $^{+30}_{-85}$ % |

2.3 Nominal Capacitance

Nominal Capacitance shall consist of three numerals in the unit of picofarad(pF).
The first and second numerals mean the significant figures, and the third numeral shall represent the number of zeros following the significant figures.
A decimal point shall be expressed by English capital R.

< Example >

| Code | Capacitance (pF) |
|---|---|
| 4R7 | 4.7 |
| 010 | 1 |
| 100 | 10 |
| 103 | 10 000 |

2.4 Capacitance Tolerance

Capacitance Tolerance shall be expressed by English capital J, K, M, N, or Z.

| Code | Tolerance |
|---|---|
| J | ± 5 % |
| K | ±10 % |
| M | ±20 % |
| N | ±30 % |
| Z | $^{+80}_{-20}$ % |

2.5 Configuration

Configuration shall be expressed by English capital A, B, NA , or KF.

| Code | Configuration |
|---|---|
| A | Axial Lead Taping (Spacing between tapes 26 mm) |
| B | Axial Lead Taping (Spacing between tapes 52mm) |
| NA | Straight Lead (Bulk Products) |
| KF | Formed Lead (Bulk Products) |

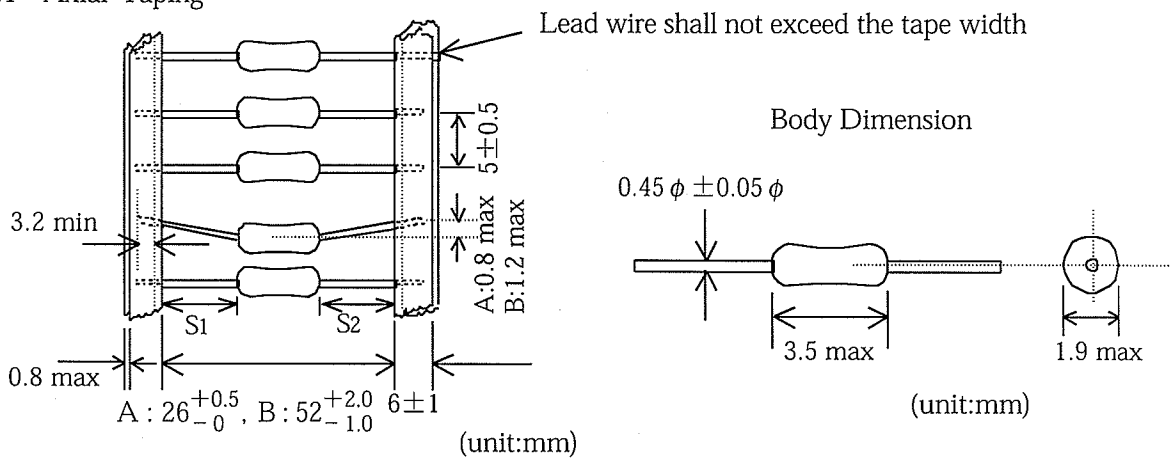
2.6 Packaging Symbol

Packaging shall be expressed by English capital B, or C.

| Code | Packaging |
|---|---|
| B | Ammo Packaging : Zig−zag folding |
| C | Bulk bagged Packaging |

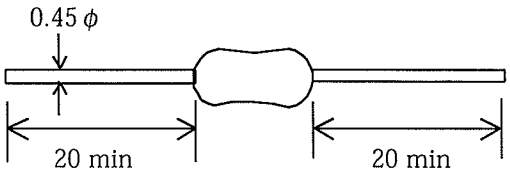
3. Axial Taping and Forming

3.1 Axial Taping



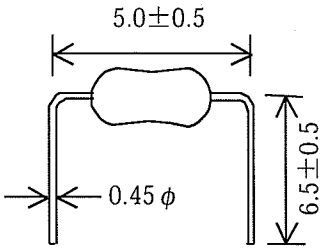

※ | S1 − S2 | ≦ 0.5

3.2 Straight Lead (NA)




(unit:mm)

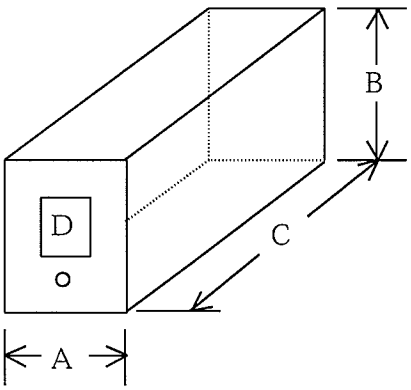
3.3 Formed Lead (KF)




(unit:mm)

4. Packaging

4.1 Ammo Packaging ; Zig−zag folding
Packaging Unit Quantity : 4 000 pcs.



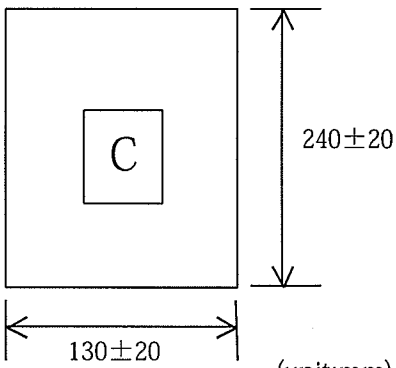

(unit:mm)

| Type Designation of Taping | A | B | C |
|---|---|---|---|
| A:26 mm | 50±5 | 70±5 | 255±5 |
| B:52mm | 70±5 | 70±5 | 255±5 |

Label•D Indication Item

① Customer part number
② Product description
③ Quantity
④ Lot number
⑤ Manufacture's name

4.2 Bulk bagged Packaging
Packaging Unit Quantity : NA=1 000 pcs.
: KF=3 000 pcs.




(unit:mm)

Label•C Indication Item

① Customer part number
② Product description
③ Quantity
④ Lot number
⑤ Manufacture's name

5. Indication

5.1 Rated Voltage

Rated voltages are indicated by the body color shown below.

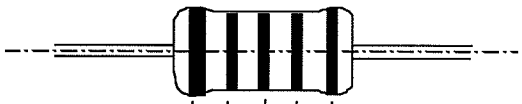
| Rated voltage (V DC) | Body color |
|---|---|
| 50 | Light greenish |
| 16, 25 | Pinknish |

5.2 Nominal Capacitance, Tolerance and Temperature Characteristics

The width of the first color zone shall be 1.5 times as that of other zones.
The line is marked on over 2/3 of circumference.

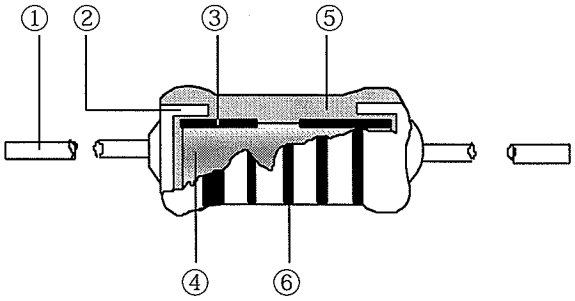
※ F temperature characteristics products has no 5th color band marking.

[Color Code Marking]



| Item | Nominal Capacitance (pF) | | | 4th Color Zone | 5th Color Zone | |
|---|---|---|---|---|---|---|
| | 1st Color Zone | 2nd Color Zone | 3rd Color Zone | | | |
| Color | (1st Figure) | (2nd Figure) | (Multiplication Factor) | (Tolerance) | (Temperature Characteristics) | |
| Black | 0 | 0 | $\times 10^0$ (1) | ±20 %(M) | CH | |
| Brown | 1 | 1 | $\times 10^1$ (10) | | | Y |
| Red | 2 | 2 | $\times 10^2$ (100) | | | |
| Orange | 3 | 3 | $\times 10^3$ (1000) | | | |
| Yellow | 4 | 4 | | | RH | |
| Green | 5 | 5 | | | | |
| Blue | 6 | 6 | | | | |
| Purple | 7 | 7 | | | UJ | |
| Gray | 8 | 8 | | ±30 %(N) | | X |
| White | 9 | 9 | | $^{+80}_{-20}$ %(Z) | SL | ※ |
| Gold | | | $\times 10^{-1}$ (0.1) | ± 5 %(J) | | |
| Silver | | | $\times 10^{-2}$ (0.01) | ±10 %(K) | | B |

6. Construction and Mechanical Performances




| No. | Components | Material |
|---|---|---|
| ① | Lead wire | Sn alloy plated soft copper wire : plating composition(Sn or Sn/Cu) |
| ② | Cap | Fe (Iron) |
| ③ | Electrode | Zn (Zinc) ※ |
| ④ | Element | Ceramic |
| ⑤ | External coating | Epoxy−based coating material |
| ⑥ | Color code | Melamine−based ink |

※ Characteristics F ⇒ Ag (Silver)

7. Operating Temperature Range
−25 ℃ to +85 ℃

8. Test conditions
Note on standard conditions : "standard conditions" referred to herein is defined as follows :
5 ℃ through 35 ℃ centigrade of temperature, 45 % through 85 % relative humidity and 86 k through 106 kPa of atmospheric pressure.
When there are questions concerning measurement results:
In order to provide correlation data, the test should be conducted under a condition of 20 ℃±2 ℃ centigrade of temperature, 60 % through 70 % of relative humidity and 86 k through 106 kPa of atmospheric pressure.
Unless otherwise specified, all the tests are conducted under the "standard condition".
If voltage is applied to Class 3 directly before measurement, thermal treatment at 150 ℃±5 ℃ for 30 minutes followed by 48±4 hours of recovery under room temperature and humidity shall be performed before the measurement.

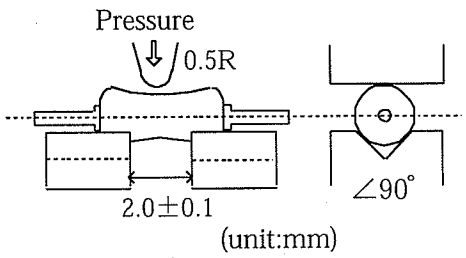
9. Rated Voltage Characteristics and Capacitors

| Rated Voltage | Class | Temperature Characteristic Symbol | Capacitance (pF) | Tolerance |
|---|---|---|---|---|
| 50(V D.C) | 1 | CH | 1.0 ～ 20 | Less than 1.8(pF) :±20 %(M)<br>2.2 ～ 8.2(pF) :±10 %(K)<br>More than 10(pF) ± 5 %(J) |
| | | RH | 1.0 ～ 18 | |
| | | UJ | 2.2 ～ 30 | |
| | | SL | 1.0 ～ 68 | |
| | 2 | B | 75 ～ 560 | ±10 %(K) |
| | 3 | B | 680 ～ 1 000 | |
| 25(V D.C) | 3 | F | 10 000. 22 000 | $^{+80}_{-20}$ %(Z) |
| 16(V D.C) | 3 | X | 1 200 ～ 6 800 | ±20 %(M), ±30 %(N) |
| | | Y | 8 200. 10 000 | |

1 0 . Capability

The capability of the Axial Lead Ceramic Capacitor is as below.

| No. | Items | | Specified Value | Test Methods and Remarks |
|---|---|---|---|---|
| 1 | Appearance and Dimension | | No abnormality | Visual and use of venire caliper and Micrometer. |
| 2 | Indication and Solvent resistivity | | Marking shall be legible after test | According to clause 8.7.4(1) of JIS C 5102.<br>Solvent type : Isopropyl alcohol<br>Duration : 30±5 seconds |
| 3 | Dielectric Strength | Between Terminals | No insulation breakdown and damage shall occur | [Class 1] Rated Voltage×3 (1~5seconds)<br>[Class 2]<br>75 pF ~ 560 pF<br>Rated Voltage×3 (1~5seconds)<br>[Class 3]<br>680 pF ~ 1 000 pF<br>Rated Voltage×1.5<br>1 200 pF ~ Y10 000 pF = 18 V<br>F10 000 pF. F22 000 pF<br>Rated Voltage×2 (1~5seconds)<br>Charging/discharging current:50 mA max.<br>(For Class 3 B{680 pF~1 000 pF}, X,Y, and F current shall be limited to 10mA) |
| | | Between Terminals and Overcoat | No abnormality | Rated Voltage×2.5 (1~5seconds)<br>Charging/discharging current : 50 mA max. |
| 4 | Insulation Resistance | Class 1,2 | 10 000 MΩmin | Application of the rated DC voltage for 60±5 seconds before reading. |
| | | Class 3 | 1 000 MΩmin | |
| 5 | Capacitance | | It shall be within specified tolerance | Measuring conditions as below.<br>Class / Frequency / Voltage<br>[1] / 1 MHz±200 kHz / 1 V±0.5 Vrms<br>[2.3] / 1 kHz±200 Hz / 1 V±0.5 Vrms<br>Characteristics F / 1 kHz±200 Hz / Under 0.1 Vrms<br>Measuring Temperature = 20 ℃<br>Measurement shall be made without application of DC bias. |
| 6 | Q (Class 1) | | 30 pF under : Q≧400+20 C※<br>33 pF over : Q≧500<br>※C=Nominal capacitance (pF)<br>(16 pF, 18 pF: Characteristics RH Q≧500) | |
| | Tan δ | Class 2 | 75 pF~ 390 pF=1.5 %max<br>470 pF~ 560 pF=2.5 %max | |
| | | Class 3 | 680 pF~1 000 pF=2.5 %max<br>1 200 pF~Y10 000 pF =2.5 %max<br>F10 000 pF. F22 000 pF =7.5 %max | |
| 7 | Temperature Characteristics | | Refer to Attachment Figure 1 to 6 | Temperature range :−25 ℃ ~ +85 ℃<br>Reference temperature : +20 ℃<br>SL temperature range : +20℃ ~ +85 ℃ |

| No. | Items | | Specified Value | Test Methods and Remarks |
|---|---|---|---|---|
| 8 | Mechanical Strength of Terminal | Tensile strength | No abnormality such as cut lead, or looseness. | One lead wire is fixed and the other is pulled axially with load of 19.6 N for1 to 5 seconds. |
| | | Bending strength | No abnormality such as cut lead, or looseness. | Suspend a mass at the end the terminal, incline the body through angle of 90° and return it to initial position.<br>Then second bend in the opposite direction shall be made.<br>Bending force : 2.45 N<br>Duration : 5 seconds |
| | | Torsion strength | No abnormality such as cut lead, or looseness. | After 5 times of 360° twist using the tool as specified in item 8.1.2(2) of JIS C 5102.<br>Time consumed in 1 twist shall be 5 seconds. |
| 9 | Vibration Endurance | | No abnormality shall result in Appearance, Capacitance, Q or Dielectric tangent, Insulation resistance and Dielectric strength shall be within the limit of specification. | According to clause 8.2 of JIS C 5102.<br>Frequency : 10 Hz to 55 Hz<br>Total Amplitude : 1.5 mm<br>Vibration frequency shall varied up and down between 10 Hz to 55 Hz in 1 minute in 3 perpendicular directions each for 2hours : total of 6 hours. |
| 10 | Shock Resistance | | No abnormality shall result in Appearance, Capacitance, Q or Dielectric tangent, Insulation resistance and Dielectric strength shall be within the limit of specification. | Capacitors are dropped freely from a height of 1 m to the floor 5 times. |
| 11 | Mechanical Strength of Main Body | | No abnormality such as damage. | Applied force : 19.6 N<br>Duration : 5 seconds<br>Speed : Shall attain to specified in 2 seconds.<br>  <br> |
| 12 | Voltage Characteristics {Applied to Characteristics B(680 pF to 1000 pF),X,Y,and F.} | | Refer to Attachment Figure 7 to 10 | After capacitance value has been measured under the condition specified in item No.5, DC biasing on the sample shall be applied by increasing gradually up to rated voltage. |

| No. | Items | Specified Value | Test Methods and Remarks |
|---|---|---|---|
| 13 | Solder Heat Resistance | No abnormality shall result in appearance and electrical performance shall be as specified in Attachment Table 1. | Lead wires are dipped in a solution of methyl alcohol containing 10 % of rosin for 2±0.5 seconds.<br>The capacitor shall be mounted to a substrate of 1.6 mm thickness with the leads inserted through its holes of 1.0 mm$\phi$, dipped in a molten solder bath with temperature as listed.<br>Temperature(℃) / Dipping time (second)<br>260± 5 / 10±1<br>350±10 / $3^{+0.5}_{-1.0}$<br>After taken out from the bath, they are left exposed in standard atmospheric conditions for 4 to 24 hours, then measurements are made. |
| 14 | Solderability | Solder shall cover more than 75 % of lead surface in peripheral direction. | Lead wires are dipped in a solution of methyl alcohol containing 25 % of rosin for 2±0.5 seconds then dipped through up to the root of main body in molten solder with temperature, 230 ℃±5 ℃ for 2±0.5 seconds, and taken out.<br>The lead wires shall satisfy the solderability conditions mentioned above when the Capacitors are used within 6 months from the time of delivery. |
| 15 | Temperature Cycle | No abnormality shall result in appearance and electrical performance shall be as specified in Attachment Table 1. | Condition for 1 cycle<br>Number of cycles : 5<br>Step / Temperature (℃) / Duration (minute)<br>1 / $-25^{+0}_{-3}$ / 30±3<br>2 / $85^{+3}_{-0}$ / 30±3<br>They are left exposed in standard atmospheric conditions for 1 hour, then measurements are made. |
| 16 | Humidity Resistance | No abnormality shall result in appearance and electrical performance shall be as specified in Attachment Table 1. | Temperature : 40 ℃±2 ℃<br>Humidity : 90 % to 95 % RH<br>Duration : $500^{+24}_{-0}$ hours<br>Recovery : They are left exposed in atmosphere of normal temperature/humidity for 1 hour and measurements are made. |

| No. | Items | Specified Value | Test Methods and Remarks |
|---|---|---|---|
| 17 | Humidity Loading | No abnormality shall result in appearance and electrical performance shall be as specified in Attachment Table 1. | Temperature : 40 ℃±2 ℃<br>Humidity : 90 % to 95 % RH<br>Duration : 500 $^{+24}_{-0}$ hours<br>Applied Voltage<br>: Rated Voltage<br>Recovery<br>: They are left exposed in atmosphere of normal temperature/humidity for 1 hour and measurements are made. For capacitors with characteristics B (680 pF to 1 000 pF), X, Y, and F they are thermal treated with temperature of 150 ℃±3 ℃ for 30 minutes and left exposed in atmosphere of normal temperature/humidity for 1 hour, then measurements are made. |
| 18 | High Temperature Loading | No abnormality shall result in appearance and electrical performance shall be as specified in Attachment Table 1. | Temperature : 85 ℃$^{+3}_{-0}$ ℃<br>Duration : 1 000 $^{+48}_{-0}$ hours<br>Applied Voltage:<br>1 pF ～ 560 pF<br>=Rated Voltage × 2<br>680 pF ～ 1000 pF<br>=Rated Voltage × 1.5<br>X 1 200 pF ～ Y10 000 pF=18 V<br>F10 000 pF・F22 000 pF<br>=Rated Voltage × 1.5<br>Recovery<br>: They are left exposed in atmosphere of normal temperature/humidity for 1 hour and measurements are made. For capacitors with characteristics B (680 pF to 1 000 pF), X, Y, and F they are thermal treated with temperature of 150 ℃±3 ℃ for 30 minutes and left exposed in atmosphere of normal temperature/humidity for 1 hour, then measurements are made. |

※ JIS Compendium for 1994.

1 1. Manufacturing location

DONG GUAN TAIYO YUDEN CO., LTD. (CHINA)
KOREA TONG YANG YUJUN CO., LTD. (KOREA)

1 2. Notice

Note 1 : The value of rated voltage applied to the capacitors' terminal shall be within the voltage range specified.
Semiconductor Capacitors are not suitable for circuits where they may be exposed to abnormal voltage loads.
※ Abnormal voltage : e.g.surge voltage, pulse voltage, static electricity, etc.

Note 2 : The capacitance of products with a high dielectric constant can decrease due to the passage of time.
Therefore, special note should be taken during design.
However, the capacitance shall recover its initial value by heat treatment (at 150 ℃ for 1 hour).

Note 3 : Flow soldering condition
Insert lead wire to substrate with t=1.6mm and hole diameter 1.0mm $\phi$、and perform soldering with solder temperature 245℃±5℃ for 3±1s.
Soldering shall be limited up to 3 times and the capacitor body shall not be immersed into the solder.

Note 4 : This component is intented for flow soldering only.
Do not use this component in reflow soldering process.

Note 5 : In case of manual soldering, please perform soldering with temperature up to 360℃ and within 3s.
The soldering iron tip shall not directly contact to the product body.

1 3. Precautions on usage

The operating conditions for the guarantee of this product are as shown in the specification.
Please note that Taiyo Yuden Co., LTD. shall not be responsible for a failure and/or abnormality which are caused by use under the conditions other than the aforesaid operating conditions.

1 4. Others

For practical use please refer to technical report "Guideline on usage of EIAJ RCR−2333 and EIAJ RCR−2331" published by EIAJ, EMAJ.
For any questions, please feel free to contact us.

Request regarding washing solvent

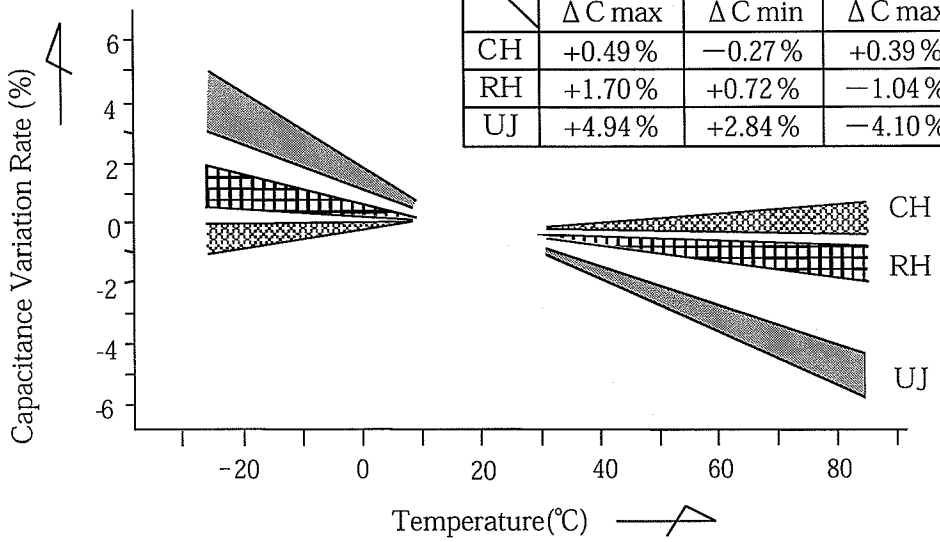
If washing is required, please use the following washing solvent.
Alcohol solvent
Isopropyl alcohol
Caution must be taken for the control and maintenance of the solvent even with the above solvent.

Note 1 : The washing solvent with chlorinated substances such as trichloroethane (containing free chlorine) shall not be used.

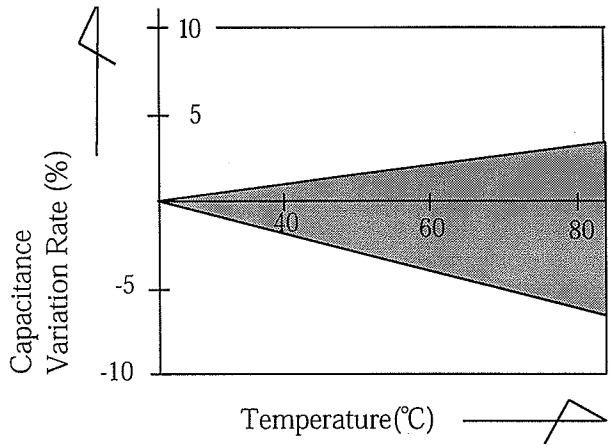
Note 2 : In case ultrasonic washing is needed, please consult with us in advance.

Attachment Figure 1
Temperature Characteristics : CH, RH, UJ

| | −25℃ | | +85℃ | |
|---|---|---|---|---|
| | ΔC max | ΔC min | ΔC max | ΔC min |
| CH | +0.49% | −0.27% | +0.39% | −0.39% |
| RH | +1.70% | +0.72% | −1.04% | −1.82% |
| UJ | +4.94% | +2.84% | −4.10% | −5.66% |

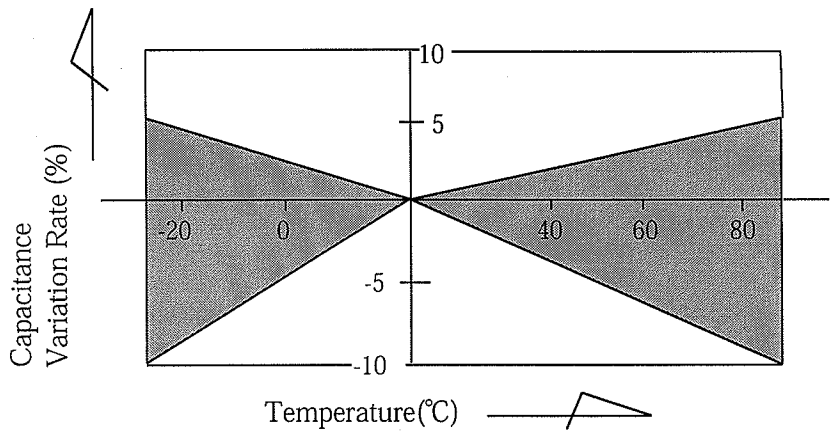



Attachment Figure 2
Temperature Characteristics : SL

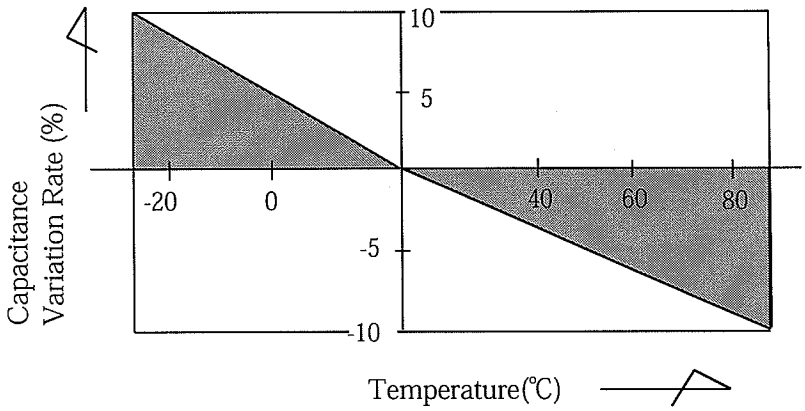



| +85℃ | |
|---|---|
| ΔC max | ΔC min |
| 2.28% | −6.50% |

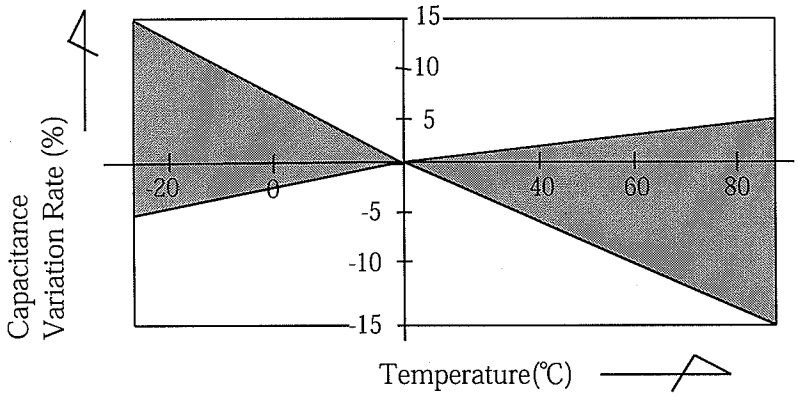
Attachment Figure 3
Temperature Characteristics : B (Class 2)



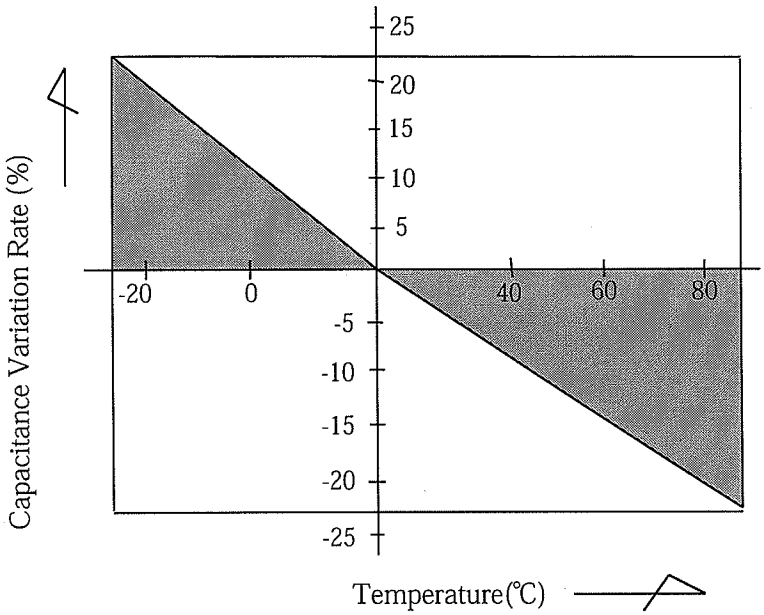
Attachment Figure 4
Temperature Characteristics : B (Class 3)




Attachment Figure 5
Temperature Characteristics : X, Y
Characteristics X (±15%)

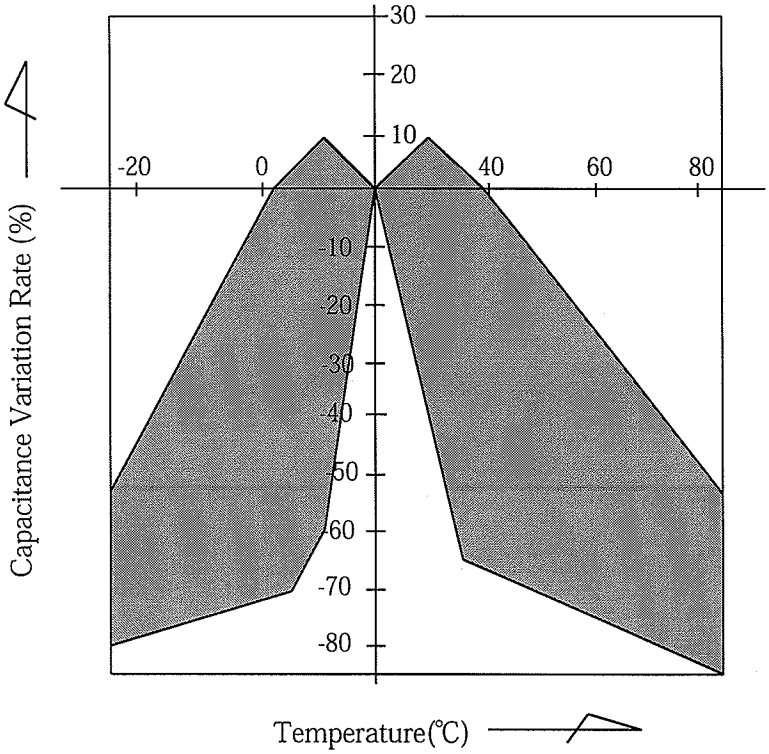



Characteristics Y (±22%)



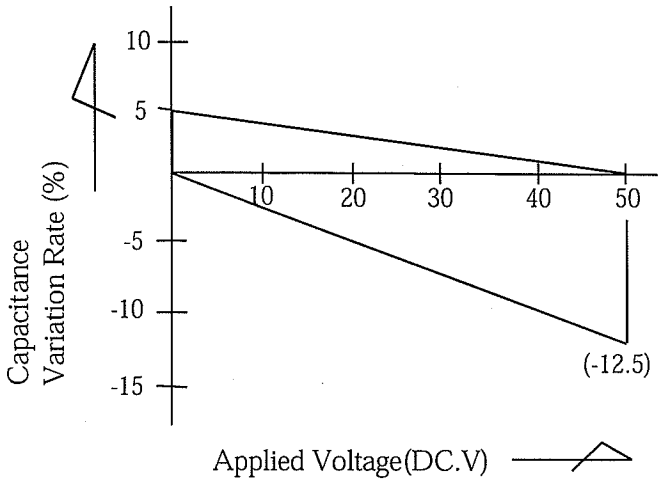
Attachment Figure 6

Temperature Characteristics : F $^{+30}_{-85}$ %

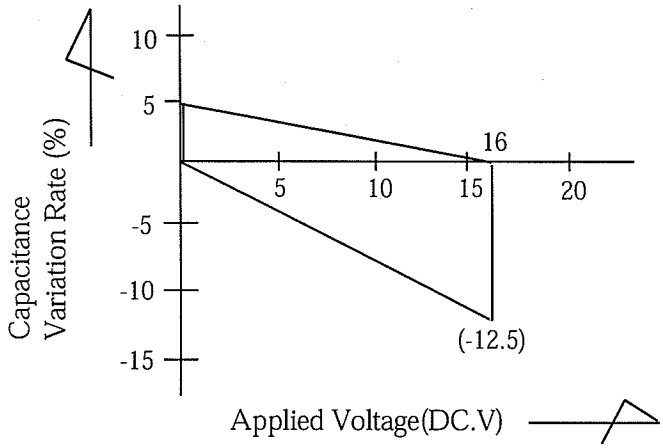



Attachment Figure 7

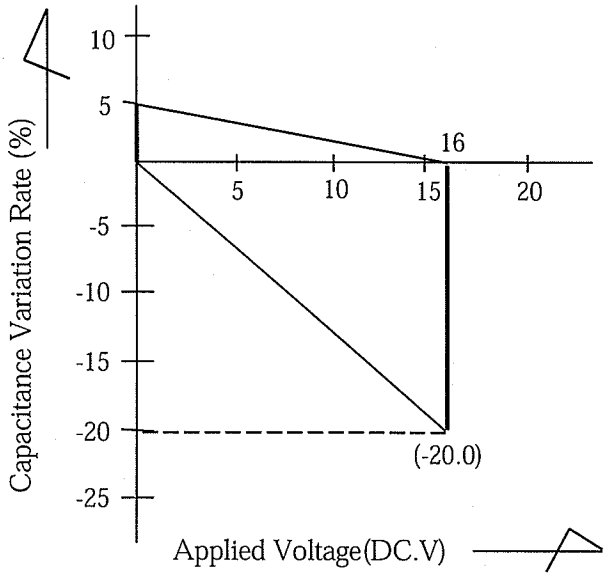
Voltage Characteristics : B (Class 3)



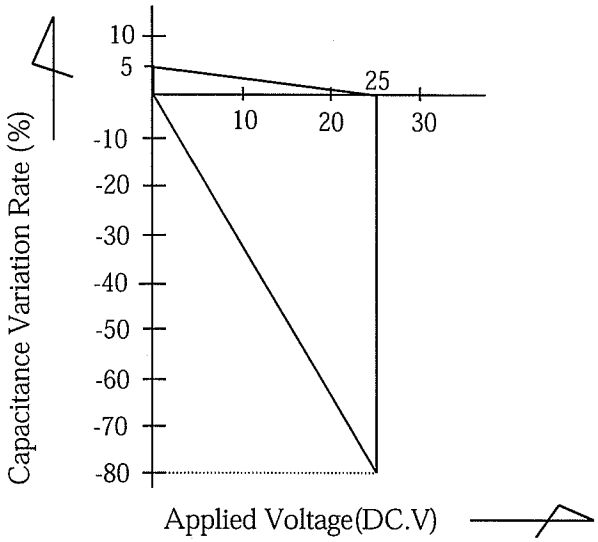
Attachment Figure 8
Voltage Characteristics : X




Attachment Figure 9
Voltage Characteristics : Y




Attachment Figure 10
Voltage Characteristics : F



# Attachment — Table 1

| Type Name | Voltage Rating | Nominal Capacitance | Capacitance Tolerance | Characteristics | Temperature-Cycle/Humidity Resistance | | | Loading under Humidity | | | Loading under High Temperature | | | Soldering Heat Resistance | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Capacitance Variation Rate | Q | Insulation Resistance (MΩ) | Capacitance Variation Rate | Q | Insulation Resistance (MΩ) | Capacitance Variation Rate | Q | Insulation Resistance (MΩ) | Capacitance Variation Rate | Q | Insulation Resistance (MΩ) |
| | DC V | pF | % | | within | more than | more than | within | more than | more than | within | more than | more than | within | more than | more than |
| UP050 □□ 010M | 50 | 1.0 | ±20 | SL CH RH | ±0.5pF | 210 | 1000 | ±0.75pF | 103 | 500 | ±0.3pF | 210 | 1000 | ±0.25pF | 420 | 10000 |
| 1R2M | | 1.2 | | | | 212 | | | 104 | | | 212 | | | 424 | |
| 1R5M | | 1.5 | | | | 215 | | | 105 | | | 215 | | | 430 | |
| 1R8M | | 1.8 | ±20 | | | 218 | | | 106 | | | 218 | | | 436 | |
| 2R2K | | 2.2 | ±10 | UJ | | 222 | | | 108 | | | 222 | | | 444 | |
| 2R7K | | 2.7 | | | | 227 | | | 109 | | | 227 | | | 454 | |
| 3R3K | | 3.3 | | | | 233 | | | 111 | | | 233 | | | 466 | |
| 3R9K | | 3.9 | | | | 239 | | | 113 | | | 239 | | | 478 | |
| 4R7K | | 4.7 | | | | 247 | | | 116 | | | 247 | | ±0.25pF | 494 | |
| 5R6K | | 5.6 | | | | 256 | | | 119 | | | 256 | | ±5% | 512 | |
| 6R8K | | 6.8 | | | | 268 | | | 123 | | | 268 | | | 536 | |
| 8R2K | | 8.2 | ±10 | | | 282 | | | 128 | | | 282 | | | 564 | |
| 100J | | 10 | ± 5 | | ±0.5pF | 300 | | ±0.75pF | 134 | | ±0.3pF | 300 | | | 600 | |
| 110J | | 11 | | | ±5% | 302 | | ±7.5% | 137 | | ±3% | 302 | | | 620 | |
| 120J | | 12 | | | | 305 | | | 140 | | | 305 | | | 640 | |
| 130J | | 13 | | | | 307 | | | 144 | | | 307 | | | 660 | |
| 150J | | 15 | | RH | | 312 | | | 150 | | | 312 | | | 700 | |
| ※ 160J | | 16 | | | | 315 | | | 154 | | | 315 | | | 720 | |
| ※ 180J | | 18 | | | | 320 | | | 160 | | | 320 | | | 760 | |
| 200J | | 20 | | CH | | 325 | | | 167 | | | 325 | | | 800 | |
| 220J | | 22 | | | | 330 | | | 174 | | | 330 | | | 840 | |
| 240J | | 24 | | | | 335 | | | 180 | | | 335 | | | 880 | |
| 270J | | 27 | | | | 343 | | | 190 | | | 343 | | | 940 | |
| 300J | | 30 | | UJ | | 350 | | | 200 | | | 350 | | | 1000 | |
| UP050 SL 330J | | 33 | | | | 250 | | | 125 | | | 250 | | | 500 | |
| 360J | | 36 | | | | | | | | | | | | | | |
| 390J | | 39 | | | | | | | | | | | | | | |
| 430J | | 43 | | | | | | | | | | | | | | |
| 470J | | 47 | | | | | | | | | | | | | | |
| 510J | | 51 | | | | | | | | | | | | | | |
| 560J | | 56 | | | | | | | | | | | | | | |
| 620J | | 62 | | | | | | | | | | | | | | |
| 680J | 50 | 68 | ±5 | SL | ±5% | 250 | 1000 | ±7.5% | 125 | 500 | ±3% | 250 | 1000 | ±5% | 500 | 10000 |
| | | | | | | | | | | | | | | | | |
| ※ RH 160J | 50 | 16 | ±5 | RH | ±5% | 250 | 1000 | ±7.5% | 125 | 500 | ±3% | 250 | 1000 | ±5% | 500 | 10000 |
| ※ RH 180J | 50 | 18 | ±5 | RH | ±5% | 250 | 1000 | ±7.5% | 125 | 500 | ±3% | 250 | 1000 | ±5% | 500 | 10000 |

# Attachment — Table 1

| Type Name | Voltage Rating | Nominal Capacitance | Capacitance Tolerance | Characteristics | Temperature-Cycle/ Humidity Resistance | | | Loading under Humidity | | | Loading under High Temperature | | | Soldering Heat Resistance | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Capacitance Variation Rate | tan δ (%) | Insulation Resistance (MΩ) | Capacitance Variation Rate | tan δ (%) | Insulation Resistance (MΩ) | Capacitance Variation Rate | tan δ (%) | Insulation Resistance (MΩ) | Capacitance Variation Rate | tan δ (%) | Insulation Resistance (MΩ) |
| | DC V | pF | % | | within | less than | more than | within | less than | more than | within | less than | more than | within | less than | more than |
| UP050 B750K | 50 | 75 | ±10 | B | ±10% | 2.5 | 1000 | ±10% | 2.5 | 500 | ±10% | 2.5 | 1000 | ±10% | 1.5 | 10000 |
| B820K | | 82 | | | | | | | | | | | | | | |
| B910K | | 91 | | | | | | | | | | | | | | |
| B101K | | 100 | | | | | | | | | | | | | | |
| B121K | | 120 | | | | | | | | | | | | | | |
| B151K | | 150 | | | | | | | | | | | | | | |
| B181K | | 180 | | | | | | | | | | | | | | |
| B221K | | 220 | | | | | | | | | | | | | | |
| B271K | | 270 | | | | | | | | | | | | | | |
| B331K | | 330 | | | | | | | | | | | | | | |
| B391K | | 390 | | | | 2.5 | | | 2.5 | | | 2.5 | | | 1.5 | |
| B471K | | 470 | | | | 4 | | | 5 | | | 4 | | | 2.5 | |
| B561K | | 560 | | | | | 1000 | | | 500 | | | 1000 | | | 10000 |
| B681K | | 680 | | | | | 500 | | | 250 | | | 500 | | | 1000 |
| B821K | | 820 | | | | | ≦ | | | ≦ | | | ≦ | | | ≦ |
| B102K | 50 | 1000 | ±10 | B | ±10% | 4 | 500 | ±10% | 5 | 250 | ±10% | 4 | 500 | ±10% | 2.5 | 1000 |
| | | | | | | | | | | | | | | | | |
| EP050 X122M,N | 16 | 1200 | ±20,30 | X | ±10% | 4 | 500 | ±10% | 5 | 250 | ±10% | 4 | 500 | ±10% | 2.5 | 1000 |
| X152M,N | | 1500 | | | | | | | | | | | | | | |
| X182M,N | | 1800 | | | | | | | | | | | | | | |
| X222M,N | | 2200 | | | | | | | | | | | | | | |
| X272M,N | | 2700 | | | | | | | | | | | | | | |
| X332M,N | | 3300 | | | | | | | | | | | | | | |
| X392M,N | | 3900 | | | | | | | | | | | | | | |
| X472M,N | | 4700 | | | | | | | | | | | | | | |
| X562M,N | | 5600 | | | | | | | | | | | | | | |
| X682M,N | | 6800 | | X | | | | | | | | | | | | |
| Y822M,N | | 8200 | | Y | | | | | | | | | | | | |
| Y103M,N | 16 | 10000 | ±20,30 | Y | ±10% | 4 | 500 | ±10% | 5 | 250 | ±10% | 4 | 500 | ±10% | 2.5 | 1000 |
| | | | | | | | | | | | | | | | | |
| TP050 F103Z | 25 | 10000 | +80 −20 | F | ±30% | 12.5 | 500 | ±30% | 12.5 | 250 | ±30% | ±10 | 500 | ±30% | 7.5 | 1000 |
| F223Z | 25 | 22000 | +80 −20 | F | ±30% | 12.5 | 500 | ±30% | 12.5 | 250 | ±30% | ±10 | 500 | ±30% | 7.5 | 1000 |